Math 110 — Week 9

Solutions

HW9 P1.

Idea. Trace can be computed in any basis, so use the one where P is simplest: P is 0 on $\operatorname{null} P$ and the identity on $\operatorname{range} P$.

$V = \operatorname{null} P \oplus \operatorname{range} P$. Take a basis $u_1, \dots, u_a$ of $\operatorname{null} P$ and $w_1, \dots, w_b$ of $\operatorname{range} P$. $Pu_i = 0$, and $w_j = Pv$ for some v gives $Pw_j = P^2 v = Pv = w_j$. So in the basis $u_1, \dots, u_a, w_1, \dots, w_b$, $\mathcal{M}(P) = \operatorname{diag}(0, \dots, 0, 1, \dots, 1)$ with b ones, and $\operatorname{tr} P = b = \dim \operatorname{range} P = \operatorname{rank} P$.

HW9 P2.

(a) **Idea.** m_T fixes the largest block for each eigenvalue and $\dim V = 5$ fixes the total size; list the ways to fill 5. The largest block for 2 has size 2 and the largest block for -1 has size 1. The eigenvalues of T are exactly the roots $2, -1$ of m_T, so with $a = \dim G(2, T) \geq 2$ and $b = \dim G(-1, T) \geq 1$, Prop. 73 gives $a + b = 5$: $(a, b) \in \{(2, 3), (3, 2), (4, 1)\}$.

(b) $\operatorname{tr} T = 2a - b$: 1, 4, 7.

(c) $\operatorname{tr} = 1$: $J_2(2) \oplus J_1(-1)^{\oplus 3}$. $\operatorname{tr} = 4$: $J_2(2) \oplus J_1(2) \oplus J_1(-1)^{\oplus 2}$. $\operatorname{tr} = 7$: $J_2(2) \oplus J_2(2) \oplus J_1(-1)$ (or $J_2(2) \oplus J_1(2) \oplus J_1(2) \oplus J_1(-1)$). Each has largest block 2 for the eigenvalue 2 and only 1×1 blocks for -1, so $m_T = (x - 2)^2(x + 1)$.

HW9 P3.

(a) **Idea.** Trace is the sum of the eigenvalues, and a nilpotent operator has only the eigenvalue 0.
N is nilpotent, so its only eigenvalue is 0 ($Nv = \lambda v$, $v \neq 0$ gives $0 = N^k v = \lambda^k v$). The trace is the sum of the eigenvalues with algebraic multiplicity, so $\operatorname{tr} N = 0$. (Or: in a Jordan basis $\mathcal{M}(N)$ has zero diagonal.)

(b) $\operatorname{diag}(1, -1)$: trace 0, and its square is I, so no power is 0.

(c) $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$: $AB = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, $BA = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, $AB - BA = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right) \neq 0$, with trace $1 - 1 = 0$.

Tr1.

Idea. $A^2 = -I$ pins down m_A; for a non-scalar 2×2 matrix, $m_A = \chi_A = x^2 - (\operatorname{tr} A)x + \det A$, so compare coefficients.

$m_A \mid x^2 + 1$. $A = cI$ would need $c^2 = -1$ with c real, impossible; so $\deg m_A = 2$ and $m_A = x^2 + 1$. For a non-scalar 2×2 matrix, $m_A = \chi_A = x^2 - (\operatorname{tr} A)x + \det A$. Comparing: $\operatorname{tr} A = 0$, $\det A = 1$.

Tr2.

Idea. The exponents in m_T are the largest blocks; fill the dimension 3.

Algebraic multiplicities $a \geq 1$ (for 1) and $b \geq 2$ (for 2) with $a + b = 3$: $a = 1$, $b = 2$. So $T \sim J_1(1) \oplus J_2(2)$, $\chi_T = (x - 1)(x - 2)^2$, $\operatorname{tr} T = 1 + 2 + 2 = 5$.

HW9 P4.

Idea. Linearity and symmetry are usually immediate; test positivity by looking for a nonzero v with $\langle v, v \rangle \leq 0$.

$\langle x, y \rangle_1$: yes. It is linear in x, symmetric, and $\langle x, x \rangle_1 = x_1^2 + 2x_2^2 > 0$ for $x \neq 0$. $\langle x, y \rangle_2$: no. $\langle e_2, e_2 \rangle_2 = -1 < 0$. $\langle x, y \rangle_3$: no. $\langle (1, -1), (1, -1) \rangle_3 = 0$ but $(1, -1) \neq 0$. (It is bilinear and symmetric; only positivity fails.)

HW9 P5.

(a) **Idea.** Linearity and symmetry come from the integral; the content is positivity: a continuous function ≥ 0 with integral 0 is 0.
Linearity in p comes from linearity of the integral; $\langle p, q\rangle = \langle q, p\rangle$; $\langle p, p\rangle = \int_{-1}^{1} p^2 \geq 0$. If $\int_{-1}^{1} p^2 = 0$, the continuous function $p^2 \geq 0$ is 0 on $[-1, 1]$, so p has infinitely many roots and $p = 0$.

(b) $\|1\|^2 = 2$, $\|x\|^2 = \int x^2 = \frac{2}{3}$, $\|x^2\|^2 = \int x^4 = \frac{2}{5}$: $\|1\| = \sqrt{2}$, $\|x\| = \sqrt{2/3}$, $\|x^2\| = \sqrt{2/5}$.

(c) $\langle 1, x\rangle = \int x = 0$ and $\langle x, x^2\rangle = \int x^3 = 0$ (odd integrands), but $\langle 1, x^2\rangle = \frac{2}{3} \neq 0$. So $1 \perp x$ and $x \perp x^2$; 1 and x^2 are not orthogonal.

HW9 P6.

Idea. Write the left side as an inner product with the all-ones vector, then apply Cauchy–Schwarz.

In $\mathbb{R}^3$ with the dot product, $a + b + c = \langle (a, b, c), (1, 1, 1)\rangle$, so $(a + b + c)^2 \leq \|(a, b, c)\|^2 \|(1, 1, 1)\|^2 = 3(a^2 + b^2 + c^2)$. Equality holds iff (a, b, c) and $(1, 1, 1)$ are dependent, i.e. iff $a = b = c$.

HW9 P7.

Idea. Expand both squared norms; the cross terms appear with opposite signs.

$\|u + v\|^2 = \langle u, u\rangle + \langle u, v\rangle + \langle v, u\rangle + \langle v, v\rangle$ and $\|u - v\|^2 = \langle u, u\rangle - \langle u, v\rangle - \langle v, u\rangle + \langle v, v\rangle$. Adding, the cross terms cancel: $2\|u\|^2 + 2\|v\|^2$. This uses only (conjugate) bilinearity, so it holds over $\mathbb{R}$ and $\mathbb{C}$.

I1.

Idea. Find two vectors whose inner product is the constant n and whose squared norms are the two factors: $u = (\sqrt{a_i})$ and $v = (1/\sqrt{a_i})$.

In $\mathbb{R}^n$, $\langle u, v\rangle = \sum \sqrt{a_i} \cdot \frac{1}{\sqrt{a_i}} = n$, $\|u\|^2 = \sum a_i$, $\|v\|^2 = \sum \frac{1}{a_i}$. Cauchy–Schwarz: $n^2 \leq \big(\sum a_i\big)\big(\sum \frac{1}{a_i}\big)$. Equality iff $u = cv$, i.e. $\sqrt{a_i} = c/\sqrt{a_i}$, i.e. all a_i equal c.